

Heredity

Science • Chapter 8 • Chapter Summary & Study Guide for Daksh

Why do you look something like your parents — but not exactly like them? Why do siblings differ even though they share the same parents? The answers lie in **heredity**: the transmission of features (traits) from one generation to the next through tiny biological instructions called **genes**. The chapter introduces **Mendel's experiments** — which laid the foundation of genetics — and explains how human sex is genetically determined.

Variation in a population

Reproduction always produces small differences. In asexual reproduction these arise mostly from copying errors. In sexual reproduction, mixing of DNA from two parents creates much larger variation. Some variations may be helpful (resistance to a disease), some harmful, and many neutral. Over many generations and a changing environment, useful variations spread — that's **evolution by natural selection**.

Mendel's experiments — laws of inheritance

Gregor Mendel (1866) experimented with garden pea plants. He chose pea because it has many clear-cut contrasting traits (tall vs dwarf, round vs wrinkled seeds, green vs yellow seeds, etc.) and is easy to self- or cross-pollinate. He counted thousands of plants — bringing mathematics to biology for the first time.

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MONOHYBRID CROSS – height of pea plant
Pure tall (TT) x Pure dwarf (tt)
      |
      F1 generation: ALL Tall (Tt)
      | (self-pollinate F1)
      F2 generation:  TT  Tt  Tt  tt
                      -----
                      3 Tall : 1 Dwarf
                      Genotype: 1 : 2 : 1
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Mendel's reasoning showed that traits are carried by units (now called **genes**) that come in alternative forms (**alleles**) — one dominant (T) and one recessive (t). Each parent passes one allele per gene. He framed two laws:

- **Law of Dominance**: when two contrasting alleles are present, only the dominant one is expressed.
- **Law of Segregation**: the two alleles of a gene separate during gamete formation, so each gamete carries only one allele.
- **Law of Independent Assortment** (from his dihybrid crosses): alleles of different genes assort independently of one another. (Demonstrated by the 9:3:3:1 ratio of round-yellow : round-green : wrinkled-yellow : wrinkled-green seeds in F_2 .)

How is sex determined in humans?

Humans have 23 pairs of chromosomes. The 23rd pair — the **sex chromosomes** — decides sex. **Females are XX; males are XY**. The mother always contributes an X. The father may pass an X (giving daughter, XX) or a Y (giving son, XY). So the **sex of the child is determined by the father's chromosome contribution** (though society has often blamed mothers — incorrect biologically). The probability of a male child is 1/2.

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Punnett square for sex determination

                Mother
                X      X
      X  |  X X  |  X X      -> daughters
Father
      Y  |  X Y  |  X Y      -> sons

ratio of sons : daughters = 1 : 1
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Acquired vs inherited traits

Only changes that happen in the **germ cells** (gametes) can be passed on. **Acquired traits** — like a loss of weight from dieting, or a learned skill — are not inherited because the DNA in germ cells is unaffected. **Inherited traits** arise from genetic differences and pass to the next generation.

★ MUST-REMEMBER POINTS

- A trait is controlled by one or more genes; each gene has alternative forms called alleles.
- Mendel's monohybrid F_2 ratio = 3 : 1 (phenotypic), 1 : 2 : 1 (genotypic).
- Mendel's dihybrid F_2 ratio = 9 : 3 : 3 : 1.
- Sex is determined by the father in humans (XY); mother always passes X.
- Acquired traits are NOT inherited; only changes in DNA of germ cells pass on.
- Variations are the raw material of evolution.

— End of Summary —